INDY INIT 2026 · PROJECT OVERVIEW

Dragonfly AI

A SCADA-native, read-only early-warning layer that spots a lift-station pump losing capacity before it fails.

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THE PROBLEM

Lift-station pumps lose capacity quietly, then fail.

14,310 gal

Morningside, Bloomington, April 2024

Equipment-failure overflow during a storm. The record does not say what failed, so we do not claim a cause.

25,000 gal

Fishers, Indiana, January 2009

Power failed and the controller’s backup battery was dead, so nobody was alerted.

Sources: Bloomington SSO dataset; IBJ, Aug 2013; IDEM Agreed Order

WHO IT HELPS

The people who keep wastewater moving.

Operators

See which pump needs inspection, with evidence in plain words, instead of finding out from an alarm.

Maintenance planners

Get a draft work order and a storm-ahead flag, so repairs happen before the weather arrives.

Small utilities

Works on on/off timestamps they already log. No new sensors and no failure labels needed.

WHAT WE BUILT

A hidden flow meter in data you already log.

1 · Measure

From pump on/off times alone, compute each pump’s capacity. Rain cancels out of the calculation.

2 · Compare

Each pump against its own history, with the sibling pump as a control group.

3 · Warn

Raise Watch, then Investigate, only when capacity falls and stays down.

Delivered through an operator UI, a read-only Ignition view, and an MCP server so any AI agent can query it.

Read-only. Never writes to a PLC or SCADA tag.

SCADA-NATIVE

We build on top of SCADA, not instead of it.

1 · Stays as it is

Data acquisition

Pumps, PLCs, RTUs and the SCADA historian keep running untouched. We replace nothing.

2 · Added on top

Dragonfly analysis

Reads the pump on/off and wet-well level data SCADA already logs. Read-only.

3 · Shown where operators work

Inside the SCADA screen

An Ignition Perspective view shows station tiles, the safety strip and the attention queue.

No new sensors, no rip-and-replace, no write path. The view only fills read-only memory tags and never writes to a device or OPC tag.

The Ignition view is a read-only add-on built on the same API as the web UI.

WHY IT WORKS

Rain moves run time. It barely moves our measure.

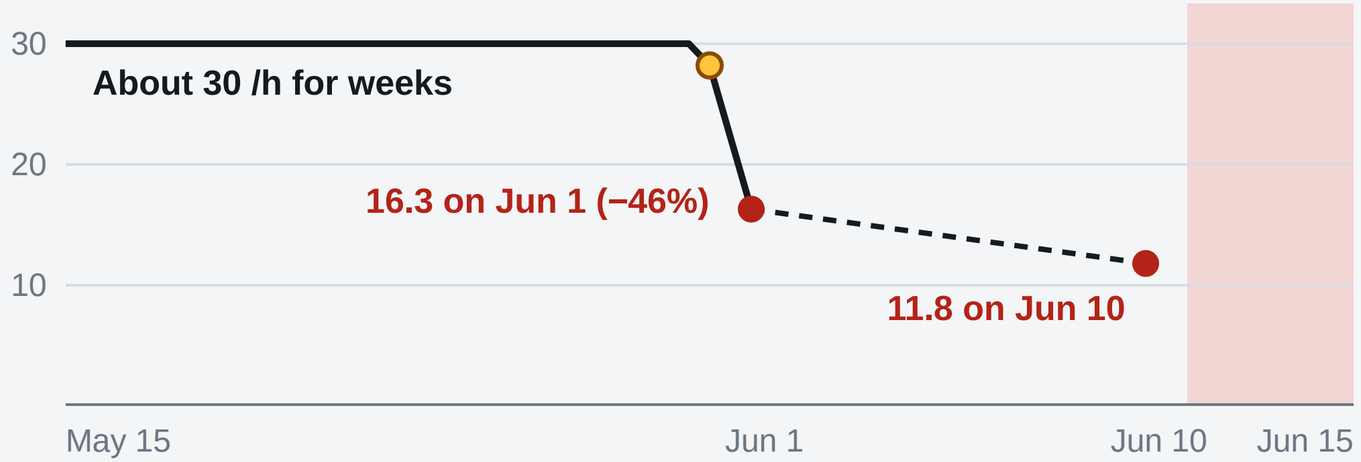


1.0 means no change on wet days. The capacity index stayed within 0.96–1.03× on 9 of 11 pumps.

Source: Wessex Water telemetry (CC BY), docs/evidence/ci_rain_check.txt. Wet day = 10 mm or more of rain.

REAL DATA

A real pump, as its capacity fell.



Shaded: pump out of service Jun 11–15. Only the readings above are plotted; the dashed line is not data.

By Jun 1, capacity was 46% below normal, ten days before the pump went out of service.

Wessex 44371 pump 1, 2022. Dev data, not a held-out test. Source: docs/evidence/ci_rain_check.txt

READING THE SCREEN

What the numbers mean.

Capacity, in “/h”

How many times this pump could empty the wet well’s working volume in an hour. Higher is stronger.

Capacity change, in %

Today against this pump’s own normal. −18.4% means 18.4% weaker than usual (example value).

Running total of shortfall

Adds up each day a pump runs below normal. One bad day fades out; a lasting drop keeps climbing.

Watch line and Investigate line

Watch at 3, Investigate at 6, and Investigate also needs 2 of the last 3 days low.

Run time: observed vs expected

Seconds per run today against what a healthy pump would take at today’s inflow (example: 212 s vs 173 s).

Evidence quality

Sufficient means 20 or more complete pump cycles that day. Otherwise it says Partial or Not enough data.

Example values come from our design mock. Rain does not change capacity, so a storm alone does not raise a flag.

HONEST BY DESIGN

It says plainly when it does not know.

Read-only

No write path to any control system. Safety alarms are never hidden.

Evidence, not guesses

Every alert shows raw timestamps, data quality and what is missing.

No failure predictions

It flags a pump to inspect. It never gives a failure date or a probability.

Simulations are labeled

Indiana incidents are record-anchored what-ifs, not real station data.

Next step: a pilot with real maintenance logs to measure true precision.

DRAGONFLY AI

Thank you.

Not a chatbot that guesses what’s wrong. A pipeline that measures each pump’s capacity from data you already log, and knows when the evidence is strong enough to interrupt an operator.

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